AAE 451 Aircraft Conceptual Design

3D-Printed Aircraft Design Course Notes

Timothée Guimara

April 30, 2026

Contents

0.1 Lecture Notes

0.1.1 Initial Vehicle Sizing by Constraint Analysis

Purpose of the Constraint Diagram

The correct sizing of an airplane depends on numerous important variables. A clearly stated mission plays a paramount role in this respect and allows the sizing to be accomplished using mathematical tools. Constraint analysis is used to assess the relative significance of selected aircraft performance parameters on the design with respect to a set of requirements.

Specifically, it allows one to understand which combinations of wing loading (W/S) and power loading (P/W) permit the aircraft to simultaneously meet dissimilar performance requirements, such as rate of climb, take-off distance, and others.

Overview of Constraint Analysis

Given mission requirements, we perform a constraint analysis using a constraint diagram. From this, we select design parameter values such as:

- Stall Speed (V_s)
- Cruise Speed (V_{cr})
- Climb rate or climb angle
- Take-off run
- Maneuver loads

Design Space: The design space is that region within which all the constraints are met. For each quantitative requirement of the mission, we find the relationship between the wing loading and the power loading that exactly satisfies that requirement.

The general form of the Constraint Equation is:

$$\left(\frac{P}{W}\right)_{Sea-Level} = f\left(\frac{W}{S}\right) \tag{1}$$

For each requirement, we find the region where the vehicle violates that requirement and indicate the forbidden region with a hash mark. The feasible region (where we want to design our aircraft) is the intersection of all allowed regions. We typically prefer smaller engines (lower P/W for cost/weight) and smaller wings (higher W/S), so the optimal point often lies at a corner of the feasible space.

1st Constraint: Sizing to Stall Speed

To satisfy the stall speed requirement, the aircraft must be able to generate enough lift to balance its weight at stall velocity V_s.

$$L = W = \frac{1}{2}\rho V_s^2 S_{ref} C_{L_{max}} \tag{2}$$

Rearranging for Wing Loading (W/S):

$$\frac{W}{S} = \frac{1}{2}\rho V_s^2 C_{L_{max}} \tag{3}$$

Where:

- W = weight
- S = wing area (S_{ref})
- ρ = air density
- V_s = stall velocity
- $C_{L_{max}}$ = maximum lift coefficient

This is a vertical line on the constraint diagram. Any wing loading greater than this value will violate the stall speed requirement.

2nd Constraint: Sizing to Cruise Speed

Assuming the aircraft is in level flight during cruise, the power required (P_{req}) must be less than or equal to the power available from the engine ($P_{avail} \cdot \eta_p$).

$$P_{req} = D \cdot V_{cr} = \frac{1}{2}\rho V_{cr}^3 S C_D \tag{4}$$

Where:

- η_p is the propeller efficiency
- C_D is the drag coefficient

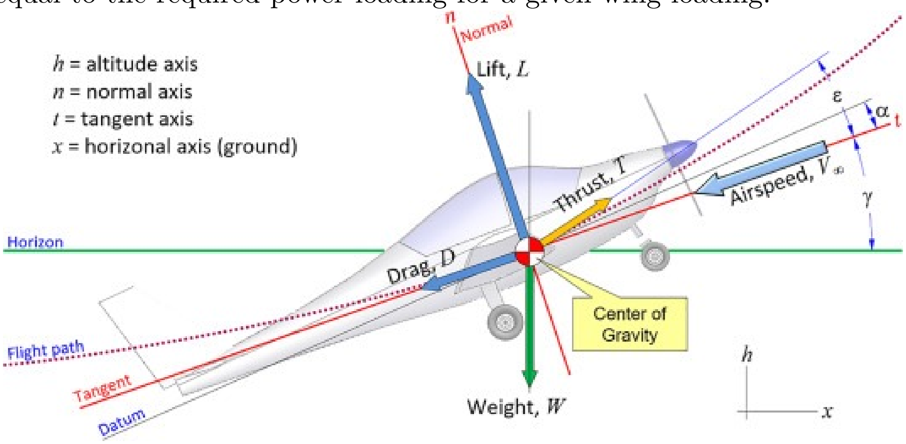
Cruise speeds for propeller-driven airplanes are typically calculated at 75 to 80 percent maximum power. Often, induced drag is small compared to profile drag, giving:

$$C_D = C_{D_0} + 0.1 C_{D_0} = 1.1 C_{D_0} \tag{5}$$

If cruise is at sea-level, the constraint equation is:

$$\frac{P}{W} \geq \frac{1.1 \rho V_{cr}^3 C_{D_0}}{2\eta_p \left(\frac{W}{S}\right)} \tag{6}$$

This equation creates a curve on the constraint diagram where power loading must be greater than or equal to the required power loading for a given wing loading.



3rd Constraint: Sizing to Climb Requirement

For an aircraft in a climb:

- Thrust direction: $T\cos(\alpha + \epsilon_T) - D = W\sin(\gamma)$
- Lift direction: $L + T\sin(\alpha + \epsilon_T) = W\cos(\gamma)$

Assuming $\alpha + \epsilon_T$ is small and $T\sin(\alpha + \epsilon_T)/W$ is small, we have:

$$L \approx W \tag{7}$$

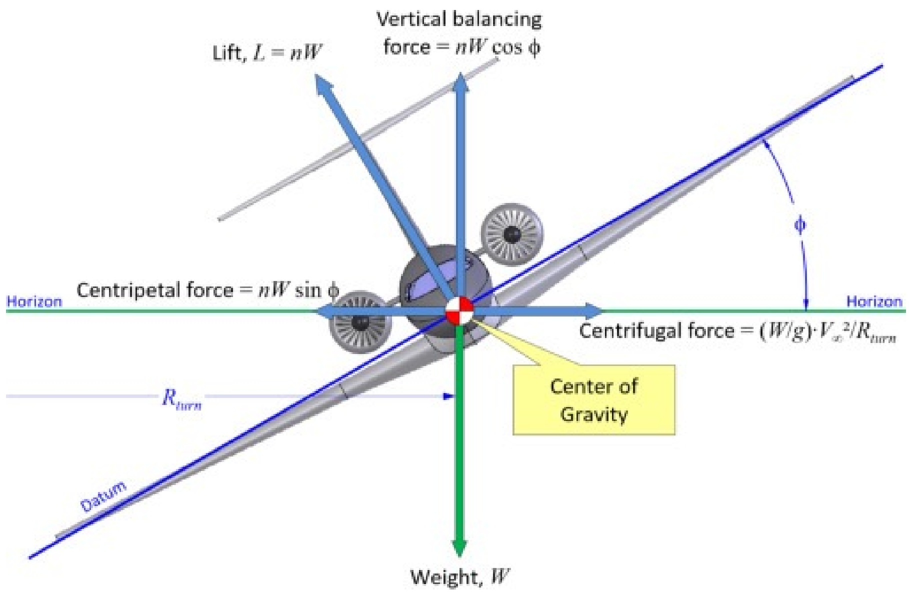
$$D = \frac{W}{L/D} \tag{8}$$

Climbing at $L/D = 0.866(L/D)_{max}$, the thrust required is:

$$T = \frac{W}{0.866(L/D)_{max}} + W\sin(\gamma) \tag{9}$$

Accounting for propeller efficiency $\eta_p = \frac{T \cdot V_{climb}}{P}$:

$$\frac{P}{W} \geq \frac{V_{climb}}{\eta_p}\left(\frac{1}{0.866(L/D)_{max}} + \sin(\gamma)\right) \tag{10}$$



4th Constraint: Sizing to Maneuver Requirement

Consider an airplane in a level constant velocity turn. The load factor is $n = 1/\cos(\phi)$, where ϕ is the bank angle.

The required thrust to maintain load factor n while banking at constant airspeed and altitude is:

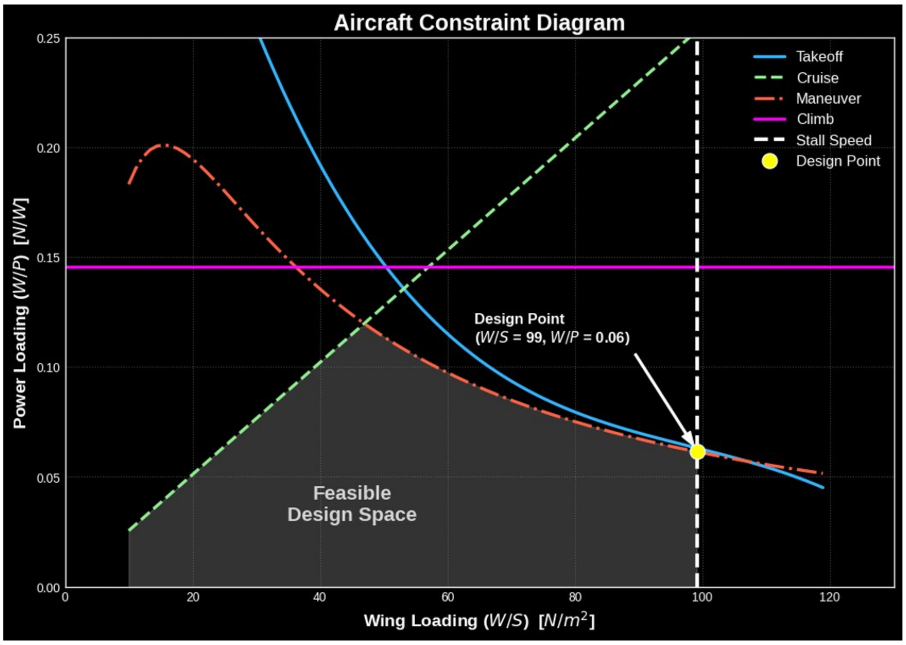
$$T = qSC_{D_0} + \frac{n^2 W^2}{qS\pi eAR} \tag{11}$$

Dividing by weight and multiplying by velocity to get power:

$$\frac{P}{W} \geq \frac{V}{\eta_p} \left(\frac{qC_{D_0}}{W/S} + \frac{n^2 (W/S)}{q\pi eAR} \right) \tag{12}$$

Where:

- $q = \frac{1}{2}\rho V^2$
- n = load factor
- AR = aspect ratio
- e = Oswald efficiency factor



5th Constraint: Take-Off

During the take-off run (S_{to}), the aircraft accelerates from $V = 0$ to $V = V_{to}$ (typically $1.1V_s$ to $1.3V_s$). The general form for the take-off constraint will depend on runway friction (μ) and aerodynamic forces during the ground roll.

Based on Sadraey's derivation (Chapter 4), the power loading constraint is:

$$\frac{W}{P} \leq \frac{\eta_P}{V_{to}} \left[\frac{1 - \exp\left(\frac{0.6 g \rho C_{D_g} S_{to}}{W/S}\right)}{\mu - \left(\mu + \frac{C_{D_g}}{C_{L_g}}\right) \exp\left(\frac{0.6 g \rho C_{D_g} S_{to}}{W/S}\right)} \right] \tag{13}$$

Where:

- S_{to} = maximum take-off distance
- C_{D_g}, C_{L_g} = drag and lift coefficients during the ground roll
- μ = runway friction coefficient
- g = gravitational acceleration

(Detailed derivation can be found in Sadraey's text book, Chapter 4).

The Master Constraint Diagram

By plotting all the above constraints (P/W vs. W/S), we identify the **Design Space**. The preferred design point is usually the one that minimizes the engine size (lowest P/W) and minimizes wing size (highest W/S) while staying within the feasible region.

> **Note:** See the Jupyter Notebook `DBF_Constraint_Diagram.ipynb` for the Python implementation and visualization of these constraints.

Reading & References

- **Gundlach:** Ch. 3, 9.4
- **Roskam (Vol. I):** Ch. 3
- **Gudmundsson:** Ch. 3
- **Sadraey:** Ch. 4

0.1.2 Initial Vehicle Weight Estimation

Our Approach

In this lecture, we focus on the initial weight estimation of electric-powered aircraft. The primary question is: **How much energy is needed to complete a design mission?**

To estimate this energy, our approach is as follows:

1. Consider how much energy is used in each mission phase.
2. Estimate the battery weight required to store that much energy.
3. Use historical data to determine the empty weight of the aircraft required to carry that battery plus any payload.

The typical mission profile involves the following phases:

1. Warmup and takeoff
2. Climb
3. Cruise (Level flight)
4. Turning flight (if applicable/maneuver)
5. Descent
6. Land and taxi

Weight Breakdown

The total aircraft weight (W) is broken down into three main components:

$$W = W_e + W_B + W_P \tag{14}$$

Where:

- W_e = Empty weight (aircraft structure, landing gear, motor, etc.; does not include battery or payload)
- W_B = Battery weight
- W_P = Payload weight

Empty Weight Prediction

Empty weight (W_e) consists of anything that cannot be removed from the aircraft without a tool. Early in the design process, when very little detail about the aircraft is known, we must use predictors or **Weight Estimating Relationships (WER)** based on historical data of similar classes of aircraft.

This historical data is usually represented as the empty weight fraction (W_e/W) plotted against the total weight (W), which typically yields a curve fit that we can use for estimation.

For a battery-powered, fixed-wing remotely controlled airplane, specific historical relations for $W_e/W = f(W)$ can be established.

Power Flow

To understand energy consumption, we must first understand the flow of power from the battery to the aircraft:

1. **Battery Output Power (P_B)**: The raw power delivered by the battery.

2. **Motor Output Power (P_m):** The mechanical power produced by the motor.

$$P_m = \eta_m \cdot P_B \tag{15}$$

Where η_m is the motor efficiency ($\sim 0.7 - 0.9$).

3. **Propeller Output Power (P_p or P_r):** The effective thrust power delivered to the aircraft.

$$P_r = P_p = \eta_p \cdot P_m \tag{16}$$

Where η_p is the propeller efficiency ($\sim 0.5 - 0.9$).

Therefore, the total power required from the battery is related to the required thrust power by:

$$P_r = \eta_p \cdot \eta_m \cdot P_B \quad \Rightarrow \quad P_B = \frac{P_r}{\eta_p \cdot \eta_m} \tag{17}$$

Energy and Mission Phases

Energy (E) is Power (P) multiplied by time (t):

$$E_{battery} = P_B \cdot t = \frac{P_r \cdot t}{\eta_p \eta_m} \tag{18}$$

We divide the mission into various flight phases and determine the energy required for each segment:

$$E_{total} = E_{to} + E_{wu} + E_{cl} + E_{lf} + E_{tf} \tag{19}$$

(Take-off, Warmup, Climb, Level Flight, Turning Flight)

The total energy required by the battery (E_B) determines the battery weight (W_B) through the **energy density** of the battery (e_B, e.g., Joules/kg).

$$W_B = \frac{E_{total}}{e_B} \tag{20}$$

Since the individual energy components are functions of the total aircraft weight (which is yet to be determined), we normalize by the unknown aircraft weight W. The problem boils down to finding the **battery weight fractions** ($W_{B,i}/W$) for each mission phase:

$$\frac{W_B}{W} = \frac{W_{B,to}}{W} + \frac{W_{B,wu}}{W} + \frac{W_{B,cl}}{W} + \frac{W_{B,lf}}{W} + \frac{W_{B,tf}}{W} \tag{21}$$

Energy Consumption in Level Flight In level flight, Lift equals Weight ($L = W$) and Thrust equals Drag ($T = D$). The thrust power required is:

$$P_r = T \cdot V = D \cdot V = \frac{W}{L/D} \cdot V \tag{22}$$

The energy required for a level flight time t_{lf} is $E_{lf} = P_r \cdot t_{lf}$. Factoring in the efficiencies and energy density, the battery weight fraction for level flight is:

$$\frac{W_{B,lf}}{W} = \frac{V \cdot t_{lf}}{\eta_p \eta_m (L/D) e_B} \tag{23}$$

Energy Consumption in Turning Flight In a turning flight with bank angle ϕ and turning radius R, the load factor $n = 1/\cos\phi$. The lift increases, and consequently, drag increases by a factor of n.

$$\frac{W_{B,tf}}{W} = \frac{V \cdot t_{tf} \cdot n}{\eta_p \eta_m (L/D) e_B} \tag{24}$$

Energy Consumption in Climbing Flight In a climb to altitude h at climb angle γ, the sum of forces yields:

$$L = W\cos\gamma \tag{25}$$

$$T = D + W\sin\gamma \tag{26}$$

Using $D = L/(L/D) = W\cos\gamma/(L/D)$, the power required is:

$$P_r = T \cdot V_{cl} = V_{cl} \cdot W \left(\frac{\cos\gamma}{L/D} + \sin\gamma \right) \tag{27}$$

For a time to climb t_{cl}, the battery weight fraction is:

$$\frac{W_{B,cl}}{W} = \frac{V_{cl} \cdot t_{cl}}{\eta_p \eta_m e_B} \left(\frac{\cos\gamma}{L/D} + \sin\gamma \right) \tag{28}$$

Energy Consumption in Warmup and Take-off During take-off for a time t_{to}, the battery output power is $P_B = \frac{P_m}{\eta_m}$. Normalizing by the airplane weight to get the inverse of the power loading (P_m/W), the battery weight fraction for take-off is:

$$\frac{W_{B,to}}{W} = \frac{t_{to}}{\eta_m \left(\frac{P_m}{W} \right)^{-1} e_B} \tag{29}$$

For warmup, a fraction (e.g., $N\%$) of the take-off energy is typically assumed:

$$\frac{W_{B,wu}}{W} = N \cdot \frac{W_{B,to}}{W} \tag{30}$$

Solution Procedure Summary

1. For a given Payload (W_P), calculate the total battery weight fraction (W_B/W) based on the mission analysis:

   $$\frac{W_B}{W} = \sum \frac{W_{B,i}}{W} \tag{31}$$

   Which gives us: $W_B = \left(\frac{W_B}{W} \right) W$.

2. We know the total weight is $W = W_e + W_B + W_P$, which can be rearranged as:

   $$W - W_e = W_B + W_P \tag{32}$$

3. Plot $(W_B + W_P)$ and $(W - W_e)$ on the same plot against total weight W (using the historical empty weight curve for W_e). The intersection of these two curves yields the **Initial Estimate of Weight**.

Suggested Reading

- **Gundlach:** Ch. 3
- **Raymer:** Ch. 3 and Ch. 20.11